CHAPTER 9 DIFFERENTIAL EQUATIONS

9.1 Modeling with Differential Equations

- **Differential Equation** An equation that contains an **unknown function** and some of its **derivatives**.

- **Exponential Growth Model**

> If the rate of growth of population is proportional to the population, then
>
> $$\frac{dP}{dt} = kP$$
>
> for some constant k.

- a reasonable model for a population of bacteria or animals under *ideal* conditions.
- Independent Variable = t
- Dependent Variable = P
- Show that $P(t) = Ae^{kt}$, for some constant A, is a **solution**.

LHS: $\frac{dP}{dt} = Ake^{kt}$

RHS: $kP = kAe^{kt}$

$\therefore P(t) = Ae^{kt}$ is a solution for $\frac{dP}{dt} = kP$

- Suppose the **initial condition** is $P(0) = 100$.

$P(0) = 100$, $P = 100$, $t = 0$

$100 = Ae^{k(0)} \Leftrightarrow 100 = A$

$\therefore P(t) = 100e^{kt}$ is solution for $\frac{dP}{dt} = kP$

- **Initial-Value Problem** A problem of finding a solution of the differential equation that satisfies the initial condition.

- **Logistic Model** (*First-Order Differential Equation*)

> The logistic equation is
>
> $$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$
>
> for some constants k and M (*carrying capacity*).

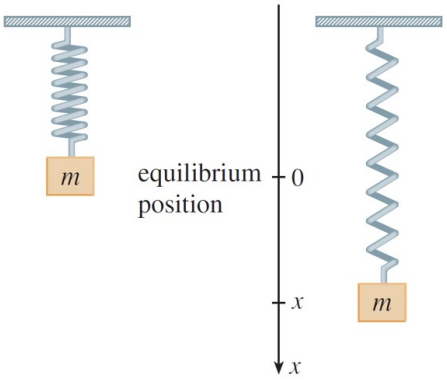
- P small:

- $P > M$:

- **Equilibrium Solutions**:

- **Mass-Spring System** (*Second-Order Differential Equation*)

> The equation of the mass-spring system is
>
> $$m\frac{d^2x}{dt^2} = -kx$$



Hooke's Law:

Newton's Second Law:

Example 9.1. Verify that $y = \sin x \cos x - \cos x$ is a solution of the differential equation $y' + (\tan x)y = \cos^2 x$, on the interval $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

$\therefore y = \sin x \cos x - \cos x$

$y' = \sin x(-\sin x) + \cos x(\cos x) + \sin x$

$= -\sin^2 x + \cos^2 x + \sin x$

$LHS = y' + \tan x\, y = -\sin^2 x + \cos^2 x + \sin x + \tan x(\sin x \cos x - \cos x)$

$= -\sin^2 x + \cos^2 x + \sin x + \frac{\sin x}{\cos x}(\sin x \cos x - \cos x)$

$= -\sin^2 x + \cos^2 x + \sin x + \sin^2 x - \sin x$

$= -\sin^2 x + 1 = \cos^2 x = RHS$

$\therefore y = \sin x \cos x - \cos x$ is a solution for

$y' + \tan x\, y = \cos^2 x$

Exercises Page 590 1, 9.

9.2 Direction Fields and Euler's Method

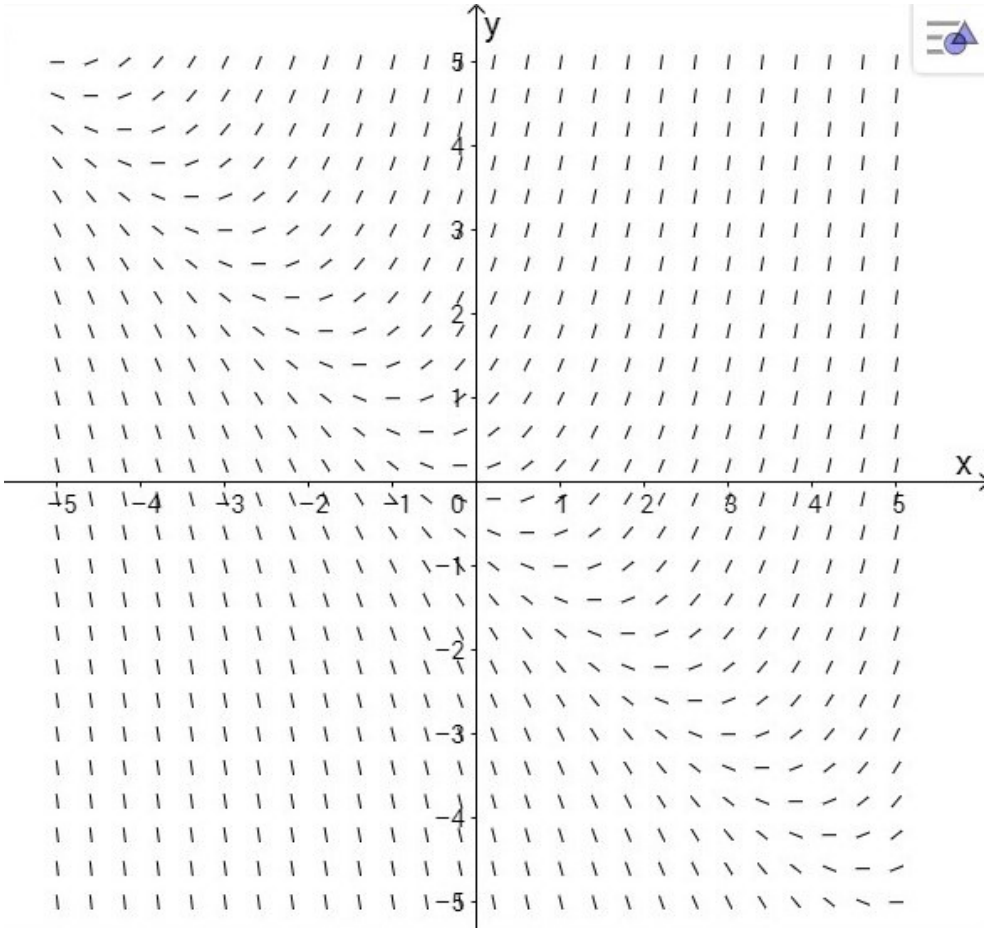
- Consider the first-order differential equation

$$y' = F(x, y)$$

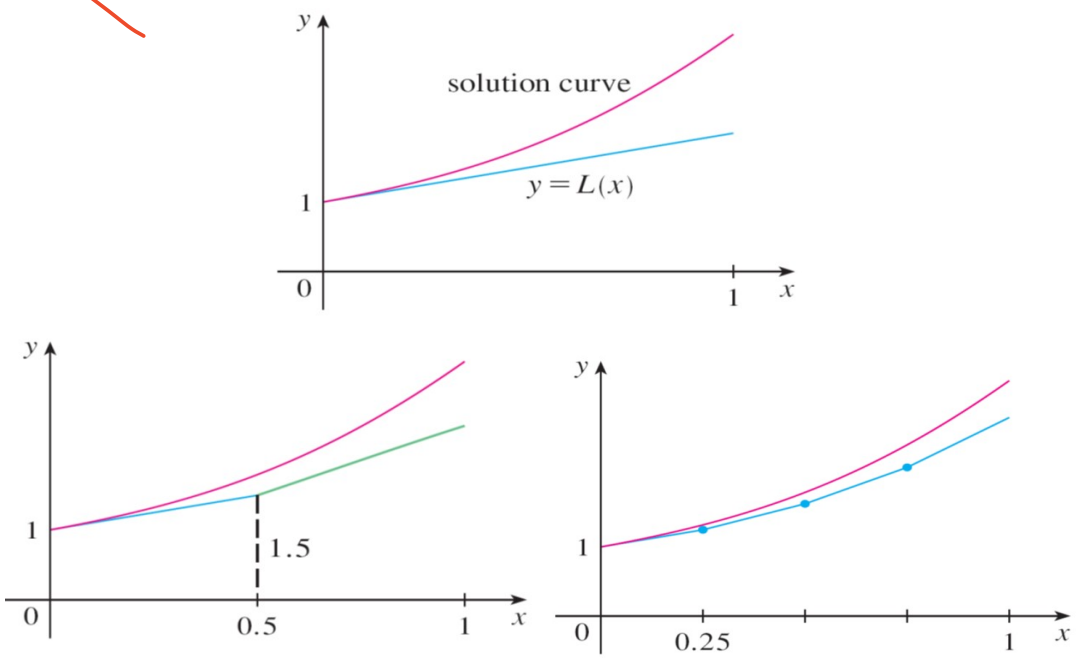
The differential equation says that the slope of a solution curve at a point (x, y) on the curve is $F(x, y)$. If we draw short lines of slope $F(x, y)$ at different points (x, y), the result is called a **direction field**. The line segments indicate the direction in which a solution curve is heading, so the direction field help us to visualize the general shape of the solution curves.

Example 9.2. A direction field for the differential equation $y' = x + y$ is shown. Sketch the graphs of the solutions that satisfy the given initial conditions.

(a) $y(0) = 0$ (b) $y(0) = 3$ (c) $y(0) = -4$

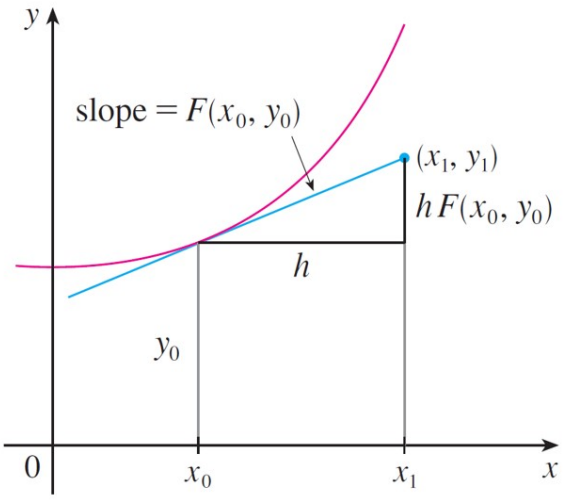

y
x
5
4
3
2
1
0
−1
−2
−3
−4
−5
−5
−4
−3
−2
−1
1
2
3
4
5

- **Euler's Method**



Approximate values for the solution of the initial-value problem $y' = F(x, y)$, $y(x_0) = y_0$, with step size h, at $x_n = x_{n-1} + h$, are

$$y_n = y_{n-1} + hF(x_{n-1}, y_{n-1})$$



Example 9.3. Use Euler's method with step size 0.1 to estimate the solution $y(0.5)$ of the initial-value problem

$$y' = x + y, \quad y(0) = 1$$

.

Exercises Page 597 1, 3 – 6, 21.

9.3 Separable Equations

Example 9.4. Solve the differential equation $\frac{dy}{dx} = \frac{y^2}{x^2}$. Then find the particular solution that satisfies the initial condition $y(1) = 2$.

$\frac{dy}{dx} = \frac{y^2}{x^2} \div y^2 \iff \frac{dy}{y^2} = \frac{dx}{x^2} \times dx$ Assume $y^2 \neq 0$

$\int \frac{dy}{y^2} = \int \frac{dx}{x^2} \iff \int y^{-2}\,dy = \int x^{-2}\,dx$

$-\frac{1}{y} = -\frac{1}{x} + C \iff \frac{1}{y} = \frac{1}{x} - C = \frac{1 - Cx}{x}$

$y = \frac{x}{1 - Cx}$ (General solution)

Example 9.5. Solve the differential equation

$$\frac{dy}{dx} = \frac{6x^2}{2y + \cos y}$$

later T

$y(1) = 2 \iff y = 2, x = 1$

$2 = \frac{1}{1 - C} \iff \frac{1}{2} = 1 - C \iff C = 1 - \frac{1}{2} = \frac{1}{2}$

The particular solution is:

$y = \frac{x}{1 - \frac{1}{2}x} \iff y = \frac{2x}{2 - x}$

Ex 9.5. Solve the differential equation

$$\frac{dy}{dx} = \frac{6x^2}{2y+\cos y}$$

$$(2y+\cos y)\,dy = 6x^2\,dx$$

$$\int (2y+\cos y)\,dy = \int 6x^2\,dx$$

$$y^2 + \sin y = 2x^3 + C \quad \text{(General solution)}$$

Example 9.6. Solve the differential equation

$$\frac{dI}{dt} + 3I = 15$$

Method I:

$$\frac{dI}{dt} = 15 - 3I \quad \xleftrightarrow[\times dt]{\div(15-3I)}$$

$$\frac{dI}{15-3I} = dt$$

$$\int \frac{dI}{15-3I} = \int dt$$

$$-\frac{1}{3}\ln|15-3I| = t + c$$

$$\xRightarrow{\times -3} \ln|15-3I| = -3t - 3c$$

$$\xLeftrightarrow{e} |15-3I| = e^{-3t-3c}$$

$$|15-3I| = e^{-3t}\, e^{-3c}$$

$$15-3I = \mp e^{-3c}\, e^{-3t}$$

$$-3I = \mp e^{-3c}\, e^{-3t} - 15$$

$$\xLeftrightarrow{\div -3} I = \pm \frac{e^{-3c}}{3} e^{-3t} + 5$$

General solution

$$I = Ae^{-3t} + 5 \quad \text{where } A = \pm\frac{e^{-3c}}{3}$$

If given intial condition

$I(0) = 15 \Leftrightarrow I = 15,\ t = 0$

$15 = Ae^{0} + 5 \Leftrightarrow \boxed{10 = A}$

The particular solution is.

$$\boxed{I = 10e^{-3t} + 5}$$

Method 2:

$\frac{dI}{dt} + 3I = 15$, $p(t) = 3$, $Q(t) = 15$

$I(t) = e^{\int p(t)\,dt} = e^{\int 3\,dt} = \boxed{e^{3t}}$

$I(t)\,I = \int Q(t)\,I(t)\,dt$

$e^{3t} I = \int 15\, e^{3t}\,dt$

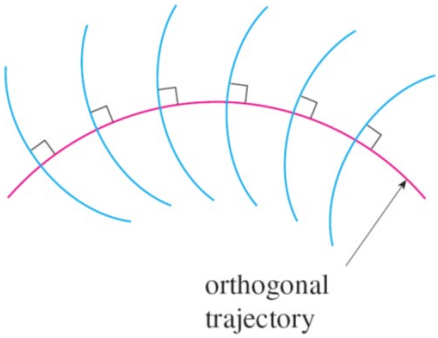
$e^{3t} I = \int 5(3)\, e^{3t}\,dt = 5e^{3t} + C$

$e^{3t} I = 5e^{3t} + C$

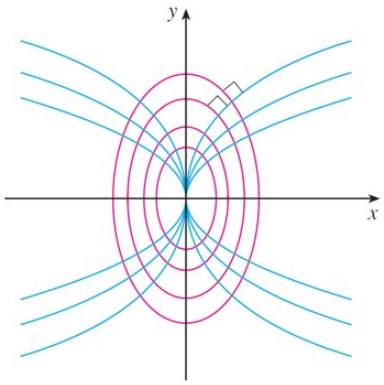
$\div e^{3t}$ ⟹ $\boxed{I = 5 + Ce^{-3t}}$ general solution

- **Orthogonal trajectory**

An **orthogonal trajectory** of a family of curves is a curve that intersects each curve of the family orthogonally.



Example 9.7. Find the orthogonal trajectory of the family of curves $x = ky^2$, where k is an arbitrary constant.



- **Mixing Problems**

Example 9.8. A tank contains 20 kg of salt dissolved in 5000 L of water. Brine that contains 0.03 kg of salt per liter of water enters the tank at a rate of 25 L/min. The solution is kept thoroughly mixed and drains from the tank at the same rate. Let $y(t)$ be the amount of salt (in kg) after t minutes.

(a) What is $y(0)$?

(b) Write $\dfrac{dy}{dt}$ in terms of y and t.

(c) Find the explicit solution $y(t)$.

(d) How much salt remains in the tank after half an hour?

Exercises Page 605 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 29, 39, 43, 45.

9.4 Models for Population Growth

- **Exponential Model**

If $P(t)$ is the value of a quantity at time t and if the rate of change of P with respect to t is proportional to its size $P(t)$, then

$$\frac{dP}{dt} = kP$$

The solution of this equation is

$$P(t) = P_0 e^{kt}$$

where $P_0 = P(0)$ is the initial condition.

- **Logistic Model**

The logistic equation is

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$

for some constants k and M. The solution of this model is

$$P(t) = \frac{M}{1 + Ae^{-kt}}, \quad A = \frac{M - P_0}{P_0}$$

where $P_0 = P(0)$ is the initial condition.

Example 9.9. Given the initial value problem

$$\frac{dP}{dt} = 0.00008P(1000 - P), \quad P(0) = 100$$

(a) This is a logistic model. What is k and M?

(b) What is the solution $P(t)$?

(c) Find the populatioin size $P(40)$.

(d) Find the populatioin size $P(80)$.

(e) When does the population reach 900?

Exercises Page 617 1, 5, 9, 21(c), 22(a).

9.5 Linear Equations

- A first-order differential equation is called **linear** if it can be put into the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where P and Q are continuous functions on a given interval.

Example 9.10. $y' + \frac{1}{x}y = 2, \; x > 0$

$$I(x) = e^{\int P(x)\,dx} = e^{\int \frac{1}{x}\,dx} = e^{\ln x} = x$$

$$x y' + x\,\frac{1}{x}\,y = 2x$$

$$x y' + y = 2x$$

$$\frac{d}{dx}(yx) = y(1) + x y' = y + x y' = \text{LHS}$$

$$\frac{d}{dx}(y\,I(x)) = 2x$$

$$\frac{d}{dx}(yx) = 2x$$

$$yx = \int 2x\,dx$$

$$yx = x^2 + C$$

$$y = x + \frac{C}{x} \quad \text{(general solution)}$$

- **Integrating Factor** $I(x) = e^{\int P(x)dx}$

$$I(x)y = \int I(x)Q(x)dx$$

Example 9.11. Solve the differential equation $\frac{dy}{dx} + 3x^2y = 6x^2$.

$P(x) = 3x^2$

$I(x) = e^{\int P(x)\,dx} = e^{\int 3x^2\,dx} = e^{x^3}$

$e^{x^3}\frac{dy}{dx} + 3x^2 e^{x^3} y = 6x^2 e^{x^3}$

$\frac{d}{dx}\left(e^{x^3} y\right) = 6x^2$

$e^{x^3} y = \int 6x^2 e^{x^3}\,dx$

let $u = x^3$

$du = 3x^2\,dx$

$e^{x^3} y = \int 2e^u\,du$

$e^{x^3} y = 2e^u + C = 2e^{x^3} + C$

$\frac{e^{x^3} y}{e^{x^3}} = 2e^{x^3} + C$

$y = 2 + \frac{C}{e^{x^3}} \Leftrightarrow y = 2 + Ce^{-x^3}$

general solution

Example 9.12. Find the solution of the initial-value problem

$$x^2y' + xy = 1, \quad x > 0, \quad y(1) = 2$$

$\div x^2 \iff y' + \frac{1}{x}y = \frac{1}{x^2}$, $p(x) = \frac{1}{x}$, $Q(x) = \frac{1}{x^2}$

$I(x) = e^{\int p(x)\,dx} = e^{\int \frac{1}{x}\,dx} = e^{\ln x} = x$

$I(x)\,y = \int I(x)\,Q(x)\,dx$

$x\,y = \int x\,\frac{1}{x^2}\,dx$

$y\,x = \int \frac{1}{x}\,dx$

$y\,x = \ln x + C$

$\div x \iff y = \frac{1}{x}\left[\ln x + C\right]$ general solution

$\because y(1) = 2 \iff y = 2\,,\ x = 1$

$2 = 1\left[\ln 1 + C\right] \iff \boxed{C = 2}$

The particular solution is:

$y = \frac{1}{x}\left[\ln x + 2\right]$

Example 9.13. Solve $y' + 2xy = 1$.

Example 9.14. A tank contains 100 L of water. A solution with a salt concentration of 0.4 kg/L is added at a rate of 5 L/min. The solution is kept mixed and is drained from the tank at a rate of 3 L/min. Let $y(t)$ be the amount of salt (in kg) after t minutes.

(a) What is $y(0)$?

(b) Show that

$$\frac{dy}{dt} = 2 - \frac{3y}{100 + 2t}$$

(c) Solve the differential equation in (b) with the initial condition in (a).

(d) Find the concentration after 20 minutes.

Exercises Page 625 5, 7, 9, 11, 13, 15, 17, 19, 33, 35.